

1 Solution — GIAM 4.1

Problem 4

1.1 Problem

What, in Logic, corresponds to the notion $\emptyset$ in Set theory?

1.2 Answer

A **contradiction** — the always-false sentence, written c .

1.3 Why

The correspondence runs through **truth sets**. Given an open sentence $P(x)$ over a universe of discourse, the text writes

$$S_P = \{x : P(x)\},$$

the collection of things that make P true. So to ask which logical notion matches $\emptyset$ is to ask: *which sentence has nothing in its truth set?*

*$S_P = \emptyset$ means no x satisfies $P(x)$ — that is, $P(x)$ is **false for every x** .*

A sentence that is false no matter what is precisely a contradiction, denoted c in the text (as in $A \wedge \neg A \cong c$).

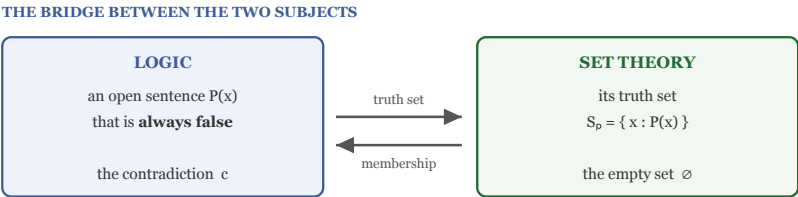
Checked on the universe $\{1, \dots, 10\}$: the always-false sentence has truth set $\emptyset$, while “ x is even” has truth set $\{2, 4, 6, 8, 10\}$ and the always-true sentence has all ten.

1.4 The Picture

THE ANSWER

$\emptyset$ corresponds to a **contradiction** — the always-false sentence **c**

A set is the collection of things making an open sentence true. Nothing makes **c** true, so its collection is empty.



THE FULL DICTIONARY

the first two rows are this exercise and the next

in Logic	in Set theory	meaning
c (contradiction)	$\emptyset$ (empty set)	always false / no members
t (tautology)	U (universal set)	always true / all members
$\neg P$	complement of S_p	not / complement
$P \wedge Q$	$S_p \cap S_t$	and / intersection
$P \vee Q$	$S_p \cup S_t$	or / union
$P \Rightarrow Q$	$S_p \subseteq S_t$	implies / subset

THE LAWS TRANSLATE TOO

Every identity law about **c** becomes an identity law about $\emptyset$, with $\wedge$ reading as $\cap$ and $\vee$ as $\cup$:

$A \wedge c \equiv c$
 $A \vee c \equiv A$
 $\neg c \equiv t$

becomes
becomes
becomes

$A \cap \emptyset = \emptyset$
 $A \cup \emptyset = A$
complement of $\emptyset = U$

All four verified on the universe $\{1, \dots, 10\}$ with $A = \{2, 3, 5, 7\}$.

MANY SENTENCES, ONE SET

On the universe $\{1, \dots, 10\}$, both " $x > 20$ " and " x is even and x is odd" have empty truth set. They are different sentences but the same set, so the correspondence is many-to-one.

Only the second is a contradiction outright; the first is merely false *on this universe*.

Figure 1.1: A diagram in four parts. The first gives the answer: the empty set corresponds to a contradiction, the always-false sentence **c**. A set is the collection of things making an open sentence true, and nothing makes **c** true, so its collection is empty. The second part draws the bridge between the two subjects as two boxes joined by arrows. On the left, in blue, headed LOGIC: an open sentence P of x that is always false, the contradiction **c**. On the right, in green, headed SET THEORY: its truth set, the set of all x with P of x , namely the empty set. An arrow labelled truth set runs left to right and an arrow labelled membership runs right to left. A table then gives the full dictionary. The contradiction **c** corresponds to the empty set, meaning always false or no members, and this row is highlighted. The tautology **t** corresponds to the universal set U , meaning always true or all members. Negation of P corresponds to the complement of the truth set of P , meaning not or complement. P and Q corresponds to the intersection of the truth sets, meaning and or intersection. P or Q corresponds to the union, meaning or or union. P implies Q corresponds to the truth set of P being a subset of the

truth set of Q , meaning implies or subset. A note says the first two rows are this exercise and the next. A band below shows that the laws translate too: every identity law about c becomes one about the empty set, with the and symbol reading as intersection and the or symbol as union. A and c equals c becomes A intersect the empty set equals the empty set; A or c equals A becomes A union the empty set equals A ; and $\text{not } c$ equals t becomes the complement of the empty set equals U . All four were verified on the universe 1 through 10 with A equal to the set 2, 3, 5, 7. A final band notes that many sentences give one set: on that universe both x greater than 20 and x is even and x is odd have empty truth set, so they are different sentences but the same set and the correspondence is many-to-one; only the second is a contradiction outright, while the first is merely false on this particular universe.

1.5 Remark — This Is the First Entry in a Dictionary

The exercise is a single line of a much larger correspondence. Logic and set theory are two presentations of the same structure, and each logical operation has a set-theoretic mirror:

Logic	Set theory	
c (contradiction)	$\emptyset$	this exercise
t (tautology)	U (universal set)	the next exercise
$\neg P$	complement of S_P	
$P \wedge Q$	$S_P \cap S_Q$	
$P \vee Q$	$S_P \cup S_Q$	
$P \implies Q$	$S_P \subseteq S_Q$	

The dictionary is worth internalising because it means **every theorem proved in one subject is a theorem in the other**. De Morgan's laws for $\wedge$ and $\vee$ become De Morgan's laws for $\cap$ and $\cup$; the distributive laws of Chapter 2 become the distributive laws for sets. Nothing needs to be reproved.

1.6 Remark — The Laws About c Become Laws About $\emptyset$

Translating the identity and domination laws of Section 2.3 through the dictionary:

logical law	set-theoretic law
$A \wedge c \cong c$	$A \cap \emptyset = \emptyset$
$A \vee c \cong A$	$A \cup \emptyset = A$
$A \wedge t \cong A$	$A \cap U = A$
$A \vee t \cong t$	$A \cup U = U$
$\neg c \cong t$	complement of $\emptyset = U$

All five were verified on the universe $\{1, \dots, 10\}$ with $A = \{2, 3, 5, 7\}$.

The two left-hand laws about c are worth reading aloud. “Anding with a contradiction gives a contradiction” becomes “intersecting with the empty set gives the empty set” — obvious in set language. “Oring with a contradiction changes nothing” becomes “unioning with the empty set changes nothing” — equally obvious. Statements that require a moment’s thought in one language can be transparent in the other, which is the practical value of having both.

1.7 Remark — The Correspondence Is Many-to-One

One caution. Different sentences can share a truth set, so the map from sentences to sets is **not** injective. On the universe $\{1, \dots, 10\}$,

$$“x > 20” \quad \text{and} \quad “x \text{ is even and } x \text{ is odd}”$$

both have empty truth set, and were confirmed to do so — yet they are different sentences.

There is a real distinction hiding here. The second is a contradiction **outright**: it is false in every universe, being of the form $A \wedge \neg A$. The first is false only *on this particular universe* — enlarge the universe to all of $\mathbb{N}$ and its truth set becomes infinite.

So the precise statement is: $\emptyset$ corresponds to *logical falsity relative to the universe of discourse*, and a genuine contradiction is the case where that falsity holds for every universe. The text’s answer, c , is the canonical representative of the class.

1.8 Remark — Why $\emptyset$ Is Unique, and What That Mirrors

In set theory there is exactly **one** empty set: if S and T both have no elements then they have the same elements vacuously, so $S = T$ by extensionality.

The logical mirror is that all contradictions are logically equivalent. $A \wedge \neg A$, $B \wedge \neg B$ and $\neg t$ are different expressions, but they have identical truth tables, so in the sense that matters they are the same object — which is why the single symbol c suffices.

The pattern recurs throughout the dictionary: uniqueness of a set-theoretic object corresponds to equivalence of the logical expressions denoting it. Extensionality for sets and logical equivalence for sentences are the same idea seen from two sides.